MATH1014 Bonus Quiz 2 Solution

Problem 1

A trough is 5 meters long, 2.5 meters wide and 4 meters deep. The vertical cross-section of the trough parallel to an end is shaped like an isosceles triangle (with height 4 meters and base, on top, of length 2.5 meters). The trough is full of water (density 1000 kg/ m^3). Find the amount of work in joules required to empty the trough by pumping the water over the top.

(Note: Use g = 9.8 m/ s^2 as the acceleration due to gravity.)

Answer:

Required Work

$$= 2\int_0^4 5g\rho A(y)ydy$$

$$= 10g\rho\int_0^4 \left(\frac{2.5}{2} - \frac{2.5}{2\times 4}\right)ydy$$

$$= 10000g\left(\int_0^4 \frac{5}{4}y - \frac{5}{16}y^2 dy\right)$$

$$= 10000\times 9.8\left(\frac{5}{8}y^2 - \frac{5}{48}y^3\Big|_0^4\right)$$

= 326666.667 J

You may directly substitute α, β, γ into the formula $\frac{4900}{3}\alpha\beta\gamma^2$ for α m long, β m wide and γ m deep trough problem in general.

Problem 2

$$\int 4x^2 \ln(1+x)dx$$

Answer: $4[\frac{x^3}{3}\ln(1+x) - \frac{x^3}{9} + \frac{x^2}{6} - \frac{x}{3} + \frac{1}{3}\ln(1+x)] + C$

Steps:

$$\int 4x^2 \ln(1+x)dx$$

$$= 4[\int \frac{x^3}{3}\ln(1+x) - \frac{1}{3}\int \frac{x^3}{1+x}dx] \quad \text{(integration by parts)}$$

$$= 4[\frac{x^3}{3}\ln(1+x) - \frac{1}{3}\int (x^2 - x + 1 - \frac{1}{1+x})dx]$$

$$= 4[\frac{x^3}{3}\ln(1+x) - \frac{1}{3}(\frac{x^3}{3} - \frac{x^2}{2} + x - \ln|1+x|)] + C$$

$$= 4[\frac{x^3}{3}\ln(1+x) - \frac{x^3}{9} + \frac{x^2}{6} - \frac{x}{3} + \frac{1}{3}\ln(1+x)] + C$$

Problem 3

For each of the following integrals indicate whether integration by substitution or integration by parts is more appropriate, or if neither method is appropriate. Do not evaluate the integral.

1. $\int x \sin x dx$ Answer: integration by parts

2. $\int \frac{x^4}{1+x^3} dx$ Answer: substitution

3. $\int x^4 e^{x^5} dx$ Answer: substitution

4. $\int x^4 \cos x^5 dx$ Answer: substitution

5. $\int \frac{1}{\sqrt{2x+1}} dx$ Answer: substitution